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Research Article

Quantifying Schedule Delay Risk in Construction Projects: A Data-Driven Approach with BIM and Probabilistic Reliability Analysis

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Progress management is an important scheme of construction management. In practical affairs, many unknown factors may lead to potential project delays and make the schedule risky. To address this issue, this paper proposes a data-driven construction progress evaluation method that employs a probabilistic reliability analysis with BIM to effectively quantify the risk of schedule delay. By identifying critical tasks and probable failure conditions in uncertain environments, this method enables project managers to take proactive measures to keep their projects on schedule. Through our case study, we have demonstrated that the proposed method is highly effective in dynamically evaluating the project progress with both accuracy and efficiency. Besides, this system offers a more efficient way for managers to gather information, evaluate progress, and identify critical tasks, and provides project managers with a data-driven understanding of the risks and uncertainties related to project schedules, enabling them to take proactive measures to minimize the potential delays and ensure successful project delivery.

1. Introduction

Progress management is a critical composing part of construction project management. Any improper management will result in project delays, cost overruns, and quality issues, which would lead to disappointment for clients, financial loss for contractors, and potential safety hazards for workers and the public [1]. Therefore, it is particularly necessary to confirm whether the project's progress is under control. According to the research in [2], despite recent advances in the information management system, the current progress management process is still mostly based on manual work, including data collection, validation, and evaluation. These works are time-consuming and error-prone. Besides, most information systems usually do not provide any advice. So, project managers are required to make decisions on their own, but some researchers showed that many construction personnel generally do not have sufficient expertise in progress management, and thus, they could hardly use complicated theoretical tools to make wiser decisions. This makes inappropriate decisions a main factor for project delays [3]. In addition, manual work relies on personal experience extensively, which goes against the standardization of project management. It is also difficult to effectively share management experience, leading to enormous costs of cultivating new employees [4]. Hence, to overcome the abovementioned shortcomings, many researchers combined the building information model (BIM) with many automated technologies to assist the project managers in better-controlling the project progress.

Building information model (BIM) is a comprehensive and data-rich digital representation of the building. It is a powerful source of detailed, granular, and diverse information that can be harnessed for effective automated progress monitoring in construction projects. BIM provides a coherent and consistent framework that combines 3D geometry and as-planned construction schedules to produce a 4D model that encompasses all relevant information for the entire construction process. This constructible and

dynamic model enables stakeholders to visualize the as-planned state of the building project at any given point in time and compare it with the actual construction state to detect any deviations. Thus, some researchers [5, 6] believed that BIM could serve as a powerful tool for monitoring the construction progress, informing decision-making processes, and enhancing the project efficiency and effectiveness.

Traditional BIM-based progress management involves integrating BIM with the project management process. Project managers use as-planned data in BIM to create 4D simulations to predict activity duration and to optimize schedules, and to monitor and control the construction process. By incorporating progress data collection, project managers can monitor the progress and detect deviations visually, allowing them to take corrective actions promptly. This approach is widely applied in the current AEC industry and research shows that by leveraging the BIM data effectively, construction teams could improve efficiency, reduce costs, and ensure timely project delivery. However, this approach requires a solid understanding of construction processes, data management, and project management principles, as mentioned before. Therefore, researchers have suggested that further enhancements can be made in the monitoring and analysis of project progress. One of the most promising improving solutions is incorporating data analysis algorithms to automate progress monitoring and analysis, thereby reducing the need for manual intervention [7].

1.1. Related Works. Capturing the current state of a construction project in an automated fashion can be performed by using a range of techniques that can be integrated with the building information model (BIM). These techniques can be classified into the following three main categories: data-driven analysis, vision-based recognition, and device-based detection.

In the pursuit of modern construction management practices, data-driven analysis has emerged as a powerful tool to gather data from different sources to generate insights into a project's current status. Majid et al. [8] have championed this technique, by exploring innovative ways to monitor and evaluate the construction projects using computer technology. They carried out an extensive survey within the Malaysian Construction Industry, leading to the creation of a prototype system, the digitalizing construction monitoring (DCM), which streamlines progress monitoring. DCM integrates construction drawings, digital images of the construction site, and planned schedules of work, providing a structured and an automated approach to real-time monitoring of the construction progress. By leveraging computer technology, DCM enables a comprehensive and a more efficient project management approach by minimizing potential delays, reducing costs, and enhancing overall construction outcomes. Another new framework for automated documentation and progress reporting of mechanical pipes in building construction projects has recently been introduced by Maalek et al. [9]. Using smartphones, this innovative approach promises to improve the quality of pipe classification and length estimation, resulting in a submillimeter pipe radius estimation accuracy. The proposed methods were subjected to rigorous laboratory and construction site evaluation, after which the framework was deemed to be exceptionally effective in improving the efficiency of mechanical pipe documentation and progress reporting in the construction industry. With its user-friendly interface, this framework is expected to streamline the entire process of mechanical pipe documentation, enabling rapid and accurate restoration of pipes and ultimately supporting the completion of building projects on time and on budget. Similarly, an approach to construction progress monitoring has recently been proposed by Ibrahimkhil et al. [10]. This method employs a simultaneous localization and mapping (SLAM) technique in combination with as-built BIM, enabling fast and accurate monitoring of the progress percentage by comparing the as-built and as-planned BIM models, aided by the programming languages Python and Dynamo. To enhance the precision of progress monitoring, the research team has incorporated the Hausdorff distance algorithm to extract pertinent objects and filter out noise from the site-scan data. By integrating these innovative technologies, the proposed method has proven to be highly efficient in monitoring the progress in large and complicated construction sites. This comprehensive framework facilitates better decision-making, enabling proactive project management and improvement of construction site performance.

Another approach to construction progress monitoring is vision-based recognition, which collects images from onsite webcams, aerial photography, or laser scanners and uses computer vision techniques to recover component-level progress information. Kropp et al. [11] have developed an innovative method for automating indoor progress monitoring through the analysis of as-built video data and as-planned BIM data. This method involves two key steps, including registering images with the 4D BIM model to accurately interpret the content and projecting relevant tasks onto the image space to determine the activity state. The method can recognize the actual state of construction activities by identifying the regions of interest in the images, increasing automation in progress monitoring, and facilitating better decision-making in project management. In addition to the Christopher Kropp et al.'s method, there are other innovative approaches to construction progress monitoring. Xue and Hou [12] proposed a novel method for monitoring the construction process of high-rise buildings using target detection and BIM registration. This involves identifying and registering unfinished building components to BIM elements, and inferring the overall construction progress based on the number of identified and registered components. Meanwhile, Han and Golparvar-Fard [13] proposed a method for automated material classification based on appearance in construction monitoring using 4D BIM and 3D point cloud models. Their approach involves aligning photos and BIM, back-projecting BIM elements onto images, and using texture and color filters for classification. This approach enables deviation detection and is a valuable tool for material classification in construction monitoring. These novel methods for construction progress monitoring offer promise in improving the efficiency and

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accuracy of project management in the construction industry. Kim et al. [14] proposed a new method for progress tracking using point cloud and BIM attributes, after reviewing existing technologies for construction-project-progress data collection and identifying their unique characteristics. Their method involves five stages and overcomes technical limitations by providing efficient and accurate progress data for construction projects. This approach enhances the ability to monitor and manage construction progress in real-time, and can assist in identifying potential issues early on in the construction process. By utilizing point cloud and BIM attributes, the proposed method can provide a more comprehensive and accurate representation of the construction progress, compared to traditional methodologies. As such, this approach has the potential to greatly benefit the construction industry in terms of optimizing project schedules, improving project efficiencies, and reducing costs. Golparvar-Fard et al. [15] presented an innovative automated approach for recognizing physical progress from construction photologs and building information models (BIMs), by incorporating a photorealistic reconstruction and fusion of the BIM into the as-built scene, followed by a machine learning scheme that enables automated detection of physical progress. Results are provided from real-life building projects showing the successful automated tracking, analysis, and visualization of progress at the schedule activity-level, even in the presence of occlusions.

Device-based detection, which involves the use of technologies such as RFID and UWB, has proven to be a dependable method for gathering precise and consistent signals that enable accurate predictions of the status of components and the estimation of construction progress. Tserng et al. [16] examined the difficulties in extending construction project control systems to job sites where paper-based processes and notebooks are not effective. The authors propose using personal digital assistants, bar-code scanning, and data entry mechanisms to streamline the flow of information in construction supply chain control systems, thereby ensuring that real-time information is shared among project participants to improve collaboration and reduce construction conflicts and delays. Integrating these technologies into construction project control systems can enhance dynamic control, information flow effectiveness, and optimize project outcomes. Navon and Sacks [17] discuss the impact of automated data collection techniques on project performance control in construction projects. The authors advocate for a fresh approach that identifies gaps between desired control functionality and current construction practices, rather than blindly following technology trends. The authors present a tool that can be used by construction companies to identify such gaps, which allows for a more targeted adoption of new technologies that can bridge gaps in the control information and improve overall project outcomes. Chin et al. [18] introduces a novel approach for managing logistics and progress control of structural steel works in high-rise building construction. The approach combines radio frequency identification and 4D CAD technology, and considers practical aspects of both the manufacturing and erection processes of the steel works. With the addition of an information system to support logistics and progress management, this innovative approach has the potential to improve efficiency and accuracy in the construction industry, particularly for high-rise building projects. Kim et al. [19] present a real-time progress management system that integrates an automated module for scheduling estimates and an RFID system for tracking construction progress, utilizing 3D-CAD and a database for accuracy. The system allows engineers to analyze real-time, actual construction progress with onsite RFID and robot systems, while maintaining an accuracy of material lists, automation of expected progress, and management of actual progress. The study demonstrates the positive impact of advanced technologies on real-time progress management in the construction industry, leading to better efficiency, accuracy, and productivity. Cho et al. [20] conducted a study to analyze the performance of an ultra-wideband (UWB) wireless network system used for mobile asset tracking at a dynamic construction site. The study involved static and dynamic error tests in different building spaces, and the researchers used regression analysis and Kalman filtering to investigate the sources of interference that affected the UWB system. The results demonstrated that different construction sites generate unique error patterns that must be taken into account when deploying UWB systems for mobile asset tracking in complex and dynamic environments.

While all three modes of monitoring can provide project progress estimation, they lack suggestions on whether timely delivery is feasible. Since the construction industry is fraught with uncertainties that are unique to its complex working environment and human resource-intensive manufacturing, making it more susceptible to delays than other industries. To address this, researchers have proposed project review techniques (PERT) [21], expanding on the classical PERT approach in which deterministic critical paths are relied upon for probability estimation. However, there is evidence that this approach may not accurately identify the most critical path. To overcome this challenge, a heuristic approach has been introduced by Soroush [22] to optimize path selection, resulting in more accurate estimates and reasonable critical path identification. Chen and Hsueh [23] have proposed a straightforward method for critical path analysis in a project network that incorporates fuzzy activity times. By using a linear programming formulation and a fuzzy number ranking method, this approach generates an optimal solution for critical path and total duration time, while also defining the most critical path and relative path degree of criticality to facilitate practical decision-making. In a complementary effort, Monhor [24] introduces a new probabilistic approach to compare path durations in a stochastic PERT with multivariate normal distribution. This concept of a probabilistically critical path is a counterpart to the deterministic critical path, thus filling the gap in the research by developing a probabilistic background theory for univariate and bivariate marginal distributions. Monhor also provides numerical results to illustrate the established probability bounds, demonstrating the practical applications of this approach. Together, these two contributions broaden

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the understanding of project review techniques and provide valuable insights into project managers across industries.

While the research conducted by these scholars provides valuable insights into assessing project progress, their algorithms suffer from a limitation in that they rely primarily on empirical data and lack a strong connection to real-time monitoring of ongoing projects as mentioned before (shown in Figure 1). Such a disconnection can lead to flawed results and ultimately, an inadequate evaluation of project success. To address these challenges, there is a need to develop a more flexible probabilistic method that utilizes progress data in conjunction with a progress monitoring system. Our research advances on this topic and provides several noteworthy advantages compared to the existing studies, which are shown in Table 1.

1.2. Objectives and Paper Structure. In this study, we attempt to develop a probabilistic method for progress evaluation to effectively quantify the risk of schedule delays. This algorithm effectively combines onsite management processes with reliability analysis techniques, providing project managers with more efficiency and comprehensibility. The four primary research objectives are listed as follows:

(1) Introduce the second-order reliability method and demonstrate how it is applied in construction progress assessment
(2) Explain why the softmax function should be taken to make the limit state function differentiable and derive the derivatives using the chain rule
(3) Propose a simplified algorithm for task completion time estimation based on collected data
(4) Verify the numerical accuracy and efficiency of our proposed method and compare it with other methods
(5) Assess the efficacy of our proposed system in enhancing the efficiency of project information exchange

The organization of the rest of this paper is summarized as follows. Section 2 provides the basic assumptions upon which our analysis is based. Section 3 then delves into the details of our mathematical model for evaluating schedule failures, emphasizing the integration of the second-order reliability method (SORM) into the schedule failure problem. This section also introduces an approximation function to ensure the target function is differentiable and derives the necessary gradients for successful SORM implementation. In Section 4, we discuss the dynamic estimation of task duration distribution using the collected data. Section 5 integrates all aspects of our approach and describes how we measured our system's success. A practical case study of a real construction project is presented in Section 6 to validate our methodology in terms of algorithmic accuracy and application efficiency. We conclude in Section 7 with a discussion of our contributions and limitations, and give our conclusion.

2. Basic Assumptions

To simplify the development of our proposed mathematical model and management system, the following four necessary assumptions have been made:

(1) Assumption 1: The project to be evaluated is assumed to be discrete and could be modeled by a critical path method (CPM). This means that the project should be composed of a sequence of well-defined tasks with small and discrete working areas. Thus, it should be noted that the linear infrastructure projects such as highways, channels, pipelines, bridges, and tunnels may not be suitable for our proposed method. Nonetheless, in the practical application of CPM, it is often a pragmatic choice to discretize the linear project by dividing it into several segments.
(2) Assumption 2: The duration of the task is assumed to be subjected to a PERT-Beta distribution, a widely used distribution in project management. This distribution allows for the estimation of task completion times based on three-point estimation and is characterized by three parameters. To simplify the implementation of updating distribution parameters, it is assumed that the estimation only reflects the completion time at the current construction rate. Although construction rate changes due to factors such as weather, material supply, mechanical condition, and th- number of personnel may occur during construction, making such a locality assumption can reduce complexity and enhance intuitiveness.
(3) Assumption 3: All task duration distributions are independent, meaning that the duration of one task does not impact the duration of another task. While this assumption might contradict to reality, since similar tasks might share the same allocated resources, construction techniques, or working environments, leading to interdependencies between tasks. However, this assumption could help us reduce the complexity of our mathematical model as well as system implementation. Moreover, independency between input variables is also required by the second-order reliability method (SORM).
(4) Assumption 4: Data collected from progress reports are reliable. As the distribution estimation is based on collected data, the reliability of the collected data directly affects the effectiveness of our analysis, and any deviation from the actual situation can result in misleading and dangerous conclusions. In order to mitigate such risks, we have implemented a verification process for each progress report submitted, which requires confirmation from the project manager prior to entry into the database. Based on this process, we consider the collected data to be reliable to a certain extent, ensuring a higher degree of accuracy for our analytical outcomes.

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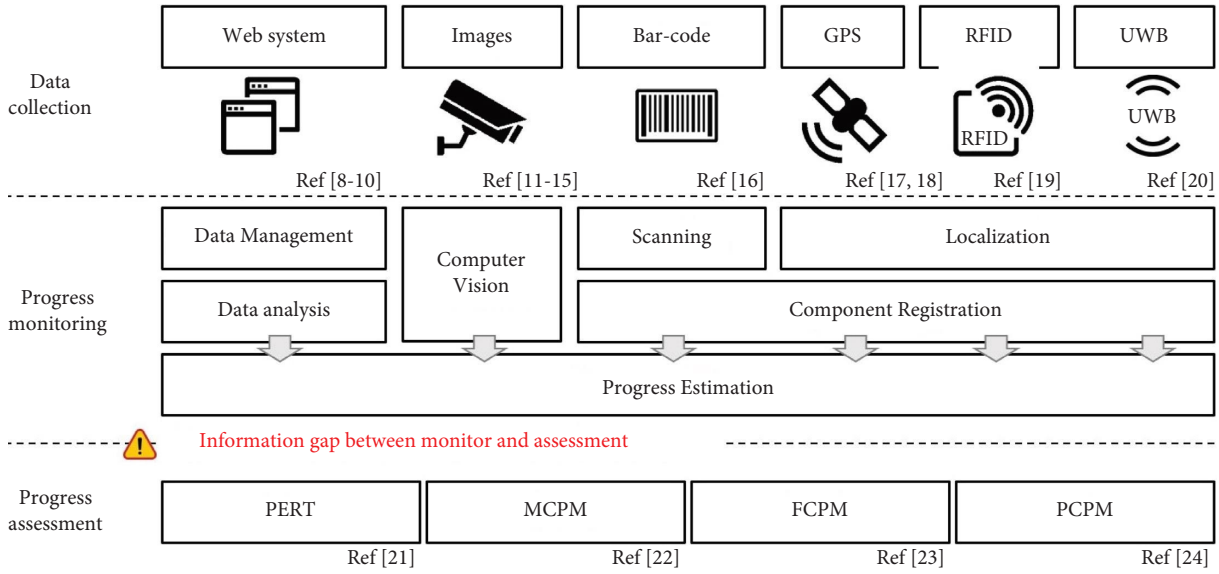


Figure 1: Current information gap between existing research studies.

3. Schedule Reliability Problem

In this section, we will begin by describing and formulating the schedule reliability problem. After that, we will introduce the second-order reliability method, which will be further approximated by applying a differentiable surrogate function. With this surrogate function in place, we can then formulate gradients using the chain rule.

3.1. Problem Description. Risk is defined by the objective uncertainty and consequences of unexpected events. One significant source of uncertainty in project management relates to the completion duration of the task. This variability in project duration increases the level of risk significantly, as it raises the possibility of exceeding the project's specified time limit. To measure the potential delay risk, we can quantify the completion time of the project as a random variable. With this approach, the likelihood of the actual project duration T_c exceeding a specified project duration T_d can be analyzed using the probabilistic methods, as shown in the following equation. By quantifying delay risk in this way, project managers can better assess and manage the uncertainties involved, leading to improved decision-making.

$$p_f = \int \mathbb{I}\left(T_c(\mathbf{D}) > T_d\right) p(\mathbf{D}) \mathrm{d}\mathbf{D}, \tag{1}$$

where $\mathbf{D} = (D_1, D_2, \ldots, D_n)^T$ is a vector collecting the duration variables D_k of every task; $T_c(\mathbf{D})$ is an implicit function transforming the durations of tasks to the completion time, which could be determined by the modified Dijkstra algorithm (MDA) introduced by Shankar and Sireesha [25], shown in the code box of Algorithm 1 and 2; $\mathbb{I}(x)$ is an indicator function that takes 1 if x is positive, otherwise, it takes 0; and $p(\mathbf{D})$ is the joint probability density function of the duration variables. Thus, our goal is to compute the delay probability and ensure it is less than a given threshold.

3.2. Second-Order Reliability Method. To solve the schedule reliability problem mentioned above, the second-order reliability method (SORM), a widely accepted reliability method, is applied to calculate the schedule reliability, which is defined as the probability of satisfying an intended function in the presence of uncertainties [26]. In SORM, the limit state function, which describes the system's behaviour under given input, is assumed to be approximated by a quadratic function near the limit state surface separating failure and nonfailure regions, as shown in the following equation:

$$g(\mathbf{u}) \approx q(\mathbf{u}) = g(\mathbf{u}^*) + \nabla(\mathbf{u}^*)(\mathbf{u} - \mathbf{u}^*)^T + \frac{1}{2}(\mathbf{u} - \mathbf{u}^*) H(\mathbf{u}^*)(\mathbf{u} - \mathbf{u}^*)^T, \tag{2}$$

where $H(\mathbf{u}^*)$ is the Hessian matrix at the most probable point (MPP, denoted by $\mathbf{u}^*$), that is,

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Table 1: Summary of related research studies on BIM-based construction progress management.

Types/methods		References	Advantage	Limitations
Data-driven analysis		[8–10]	Cheap and easy to use	Time-consuming and human-intensive
Vision-based recognition		[11–15]	Portable, low cost, and automatic	Sensitivity to lighting conditions and high complexity
Device-based detection	Bar-code	[16]	Low cost and simple	Easily damaged
	GPS	[17, 18]	Strong adaptability	Low precision and outdoor limitation
	RFID	[19]	No need for light	Error-prone in the presence of metals and liquids
	UWB	[20]	Reliable, longer read ranges, and 3D positioning	High cost and no portable device
PERT	MCPM	[22]	Fast and considerably accurate	Unable to identify most probable failure condition
	FCPM	[23]	Fast and simple to use	Potential invalid operation
	PCPM	[24]	Accurate and flexible	Time-consuming
Our method			Easy to use and low cost, bind to management processes, efficient work after setting up, dynamic assessment, and fast and high accuracy	Require few manual work and requires preprocessing

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(1) **for** i **in** Topo_Sort (Nodes)
(2) **if**Node[i].hasPredecessor()
(3) $E_S^{(i)} = \max_{j \in \text{Node}[i].\text{predecessors}} E_F^{(j)}$
(4) **else**
(5) $E_S^{(i)} = 0$
(6) $E_F^{(i)} = E_S^{(i)} + D^{(i)}$
(7) $T_e = \max_{i \in \text{Nodes}} E_F^{(i)}$
(8) **return** T_e

Algorithm 1: MDA forward algorithm.

(1) **for** i **in** Topo_Sort (Nodes).reverse()
(2) **if**Node[j].hasSuccessor()
(3) $L_F^{(i)} = \min_{j \in \text{Node}[i].\text{successors}} L_S^{(j)}$
(4) **else**
(5) $L_F^{(i)} = T_e$
(6) $L_S^{(i)} = L_F^{(i)} - D^{(i)}$
(7) $F^{(i)} = L_F^{(i)} - E_F^{(i)}$

Algorithm 2: MDA backward algorithm.

$$H(\mathbf{u}^*) = \begin{bmatrix} \frac{\partial^2 g}{\partial \mathbf{u}_1^2} & \cdots & \frac{\partial^2 g}{\partial \mathbf{u}_1 \partial \mathbf{u}_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 g}{\partial \mathbf{u}_n \partial \mathbf{u}_1} & \cdots & \frac{\partial^2 g}{\partial \mathbf{u}_n^2} \end{bmatrix}. \tag{3}$$

In Figure 2, a comparison is shown between the linear and nonlinear limit state function approximations with the same MPP. The shaded regions in the figure show the failure areas of each respective function. An observation from this comparison is that the failure probability of the nonlinear limit state function is expected to be lower than the linear limit state function. By fitting a quadratic response function to the limit state surface, SORM can determine the probability of failure to a high degree of accuracy. So, when there is a high degree of nonlinearity involved, the second-order reliability method (SORM) is preferable.

The second-order reliability method (SORM) utilizes a gradient-based Newton iterative search to locate the most probable point (MPP), which is represented by $\mathbf{u}^*$, within the failure domain. The failure boundary is then expanded around this point to obtain an approximate numerical result. The commonly used iterative MPP search algorithm is shown in Algorithm 3.

Once the most probable point (MPP) has been discovered, the approximation function can be simplified as follows:

$$q(\mathbf{u}) = \mathbf{u}_n - \left(\beta + \frac{1}{2}\mathbf{u}'^T H(\mathbf{u}^*)\mathbf{u}'\right), \tag{4}$$

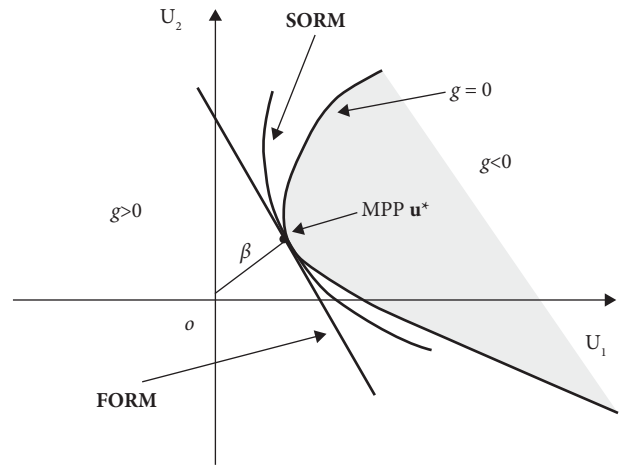


Figure 2: Comparison of FORM and SORM.

when β is large enough, the probability of failure can be approximated using Breitung's asymptotic solution [27] as

$$p_f = P(g(\mathbf{x}) < 0) = \Phi(-\beta) \prod_{i=1}^{n-1} (1 + \beta\kappa_i)^{1/2}, \tag{5}$$

where κ_i denotes the i-th main curvature of the performance function $g(\mathbf{u})$ at the design point.

Despite SORM providing a more precise approximation of the performance function compared to FORM, it has the disadvantage of requiring the computation of the second-order derivative which can be time-consuming. Hence, a second-order reliability method with first-order efficiency (SORM-FOE) is applied in our research. This method was first proposed by Zhang and Du [28], and employs a quadratic approximation function that enables SORM-FOE to

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(1) initialize $\mathbf{u}_0$
(2) **for** $k = 0 \longrightarrow$ max_iters
(3) compute $g_{\mathbf{u}}(\mathbf{u}_k)$ and $\nabla_g(\mathbf{u}_k)$
(4) $\mathbf{u}_{k+1} = ([\nabla_g^T(\mathbf{u}_k) \cdot \mathbf{u}_k - g_{\mathbf{u}}(\mathbf{u}_k)]\nabla_g(\mathbf{u}_k))/(\nabla_g^T(\mathbf{u}_k)\nabla_g(\mathbf{u}_k))$
(5) **if** $\|\mathbf{u}_{k+1} - \mathbf{u}_k\|_2/\|\mathbf{u}_k\|_2 \le \epsilon$ **then**
(6) **break**
(7) $\mathbf{u}^* = \mathbf{u}_{k+1}$
(8) **return** $\mathbf{u}^*$

Algorithm 3: Iterative MPP search.

be faster than traditional SORM. This approach helps overcome the computational time constraints associated with computing the second-order derivative, thus facilitating more efficient and accurate estimation of the performance function.

3.3. Approximation and Gradient Calculation

3.3.1. Softmax Approximation. As we can see from the pseudocode of MDA, the total duration of a schedule network could be formulated in the nested max-plus function. However, the max function is indifferentiable, thus making numerical solutions difficult to be handled by SORM. To find a surrogate function, the softmax function is applied. It is a generalization of the logistic function that convert an N-dimensional vector $\mathbf{x}$ of arbitrary real values to an N-dimensional weight vector $\sigma(\mathbf{x})$ of real values in the range $(0, 1)$ that add up to 1. The definition of weight is given by

$$\sigma(\mathbf{x})_j = \frac{e^{\mathbf{x}_j}}{\sum e^{\mathbf{x}_i}}, \text{for } j = 1, \ldots, N. \tag{6}$$

Thus, our softmax function could be derived as the weighted average value of the input vector as

$$\text{softmax}(\mathbf{x}) = \frac{\sum \mathbf{x}_i e^{\mathbf{x}_i}}{\sum e^{\mathbf{x}_i}}. \tag{7}$$

This approximation function is differentiable; however, it will also cause a systematic error. Since any number in the array is not greater than the maximum number in the array, in most cases, the weighted average is less than the real maximum. This fact could be easily explained by the following equation:

$$\text{softmax}(\mathbf{x}) = \frac{\sum \mathbf{x}_i e^{\mathbf{x}_i}}{\sum e^{\mathbf{x}_i}} \le \frac{\sum \max(\mathbf{x}) e^{\mathbf{x}_i}}{\sum e^{\mathbf{x}_i}} = \max(\mathbf{x}). \tag{8}$$

Therefore, when using such an approximation, we need to be aware that the final result will be systematically smaller than the real result. This will be discussed in a later section.

3.3.2. Gradient Backpropagation in CPM. The schedule reliability problem involves input variables in the form of activity durations, denoted as D_i. To enable the usage of SORM, it becomes necessary to derive the gradients with respect to the input variables, i.e., $\partial T_c(\mathbf{D})/\partial D_i$. However, the elimination of other variables poses a challenge, particularly in the case of nested functions. Thankfully, with the aid of the two computation units and a gradient backpropagation framework, the computation of such gradients becomes a straightforward task.

The first unit is defined by the inherent definition of a task completion duration, which is $E_i = S_i + D_i$. By computing the partial derivative of the output of this unit, which is the ending time E_i, with respect to the inputs, specifically, the starting time S_i and the task duration D_i, the equation is obtained as

$$\begin{aligned} \frac{\partial E_i}{\partial S_i} &= 1, \\ \frac{\partial E_i}{\partial D_i} &= 1. \end{aligned} \tag{9}$$

The second unit is defined by the work flow dependency, i.e., $S_i = \max_{j \in \text{Node}[i].\text{predecessors}} E_j$. To obtain the partial gradient of this unit, the max function in the constraint can be substituted with a softmax function mentioned above, and the gradients could be derived, as shown in the following equation:

$$\begin{aligned} S_i &= \text{softmax}_{j \in \text{Node}[i].\text{predecessors}} E_j = \frac{\sum_{j \in \text{Node}[i].\text{predecessors}} E_j e^{E_j}}{\sum_{j \in \text{Node}[i].\text{predecessors}} e^{E_j}}, \\ \frac{\partial S_i}{\partial E_j} &= \left(S_i - E_j - 1\right) \frac{e^{E_j}}{\sum_{k \in \text{Node}[i].\text{predecessors}} e^{E_k}}. \end{aligned} \tag{10}$$

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Based on equations (9) and (10), we could compute all the partial gradients on duration **D** we need in this project using gradient backpropagation.

This process is shown in Figure 3, and the pseudocode is given in the code box of Algorithm 4.

4. Data-Driven Distribution Updating

This section proposes a duration distribution estimation method based on historic data, by utilizing the PERT-Beta distribution as a probabilistic model. Three assumptions are made to simplify the problem.

4.1. PERT-Beta Distribution. The PERT-Beta distribution is a widely used probability distribution to estimate the duration of activities in project management [29, 30]. It combines the probabilistic evaluation and review technique (PERT) with the beta distribution, resulting in a more flexible and customizable model. PERT-Beta distribution requires the following three parameters: most likely duration b, optimistic duration a, and pessimistic duration c of the activity. The density function of PERT-Beta distribution is given by the following equation and is shown in Figure 4.

$$p(t) = \frac{1}{B(\alpha,\beta)} \frac{(t-a)^{\alpha-1}(c-t)^{\beta-1}}{(c-a)^{\alpha+\beta-1}}, \tag{11}$$

where the two shape parameters in the beta function are determined by

$$\alpha = \frac{4b+c-5a}{c-a} = 1+4\frac{b-a}{c-a}, \beta = \frac{5c-a-4b}{c-a} = 5-4\frac{b-a}{c-a}. \tag{12}$$

The relative positions of these three parameters affect the shape of the distribution. The location of the most likely outcome reflects the mode of the distribution. The spread of the distribution is influenced by the difference between the optimistic and pessimistic outcomes. When the distance between the optimistic and pessimistic outcomes is large, the distribution becomes wider and more spread out. In contrast, when the differences are smaller, the distribution becomes narrower and more concentrated around the most likely outcome, as shown in Figure 4.

4.2. Approximated Parameter Estimation. Parameter estimation of the PERT-Beta distribution is challenging as a result of its complex form, and mathematicians are still researching analytical solutions [31]. However, from an engineering application perspective, the problem can be simplified effectively. First, while the construction rate typically changes, the estimation of completion time can be accomplished by assessing the current construction rate and providing the distribution of completion time under the current rate estimate. Second, as tasks progress, the uncertainty of completion time decreases, and it can be assumed that the uncertainty is linearly reduced. Finally, the shape parameters of the beta distribution reflect the tendency of the project managers towards risk; and this tendency is assumed to be unaffected by the data. With these assumptions, an effective approximate solution can be developed.

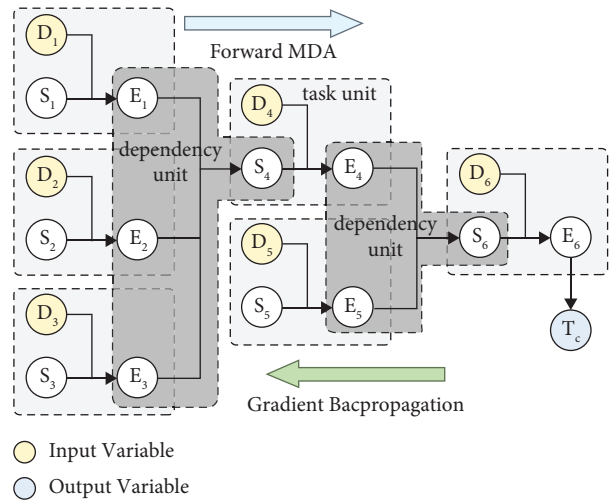


Figure 3: Basic blocks of gradient backpropagation.

To estimate the most likely duration, a customized locally weighted linear regression is applied. This approach differs from the traditional linear regression as it ensures that the regression line passes through the latest collected data points. In addition, it assigns higher fitting weights to the new data to accurately describe the current construction rate. The mathematical formulation for this method is as follows:

$$r(t) = k(t - t_*) + r_*, \tag{13}$$

where $r(t)$ is the remaining progress estimation at a time t, k is the estimation of the current construction rate, t_* is the latest report time, and r_* is the current reported remaining progress. Notice that only k need to be estimated. By combining equation (14) and the weighted sum of square loss, the loss function of our model is derived as follows:

$$\mathscr{L}(k) = \sum_{i=1}^{N} w_i \cdot (k(t_i - t_*) + r_* - r_i)^2, \tag{14}$$

where w_i is a locality weight function $w_i = \exp(-(t_i - t_*)^2/b^2)$ and b represents the locality bandwidth. A higher bandwidth value will result in an estimation that is reflective of the global view, whereas a lower value will shift the focus to the local perspective.

By solving the minimization of loss function $\mathscr{L}(k)$, we will have the best-fitted construction rate k^* as follows:

$$\frac{\partial \mathscr{L}(k)}{\partial k} = \sum_{i=1}^{N} 2w_i(k(t_i - t_*) + r_* - r_i)(t_i - t_*) = 0,$$
$$k^* = \frac{\sum_{i=1}^{N} w_i(t_i - t_*)(r_i - r_*)}{\sum_{i=1}^{N} w_i(t_i - t_*)^2}. \tag{15}$$

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```
(1) initialize Gradients(n) = 1
(2) foreach n in Topo_sort(Nodes).reverse()
(3) g0 = Gradients[n]
(4) S = Start[n]
(5) E = End[n.predecessors]
(6) g1 = (S − E − 1)exp(E)/∑ exp(E)
(7)     Gradients[n.predecessors] = g0 × g1
```

Algorithm 4: Gradients BP algorithm in CPM.

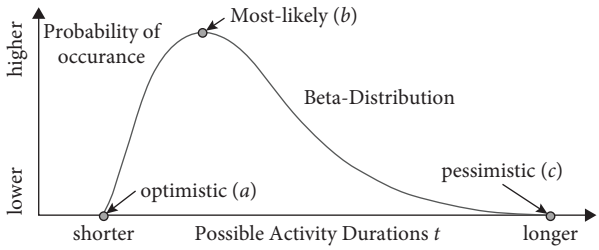


Figure 4: Typical density functions of the PERT-Beta distribution.

By substituting the estimated construction rate into the regression curve, we can obtain the most likely expected completion time b_t as follows:

$$b_t = -\frac{k^*}{c} + t_*. \tag{16}$$

To determine the duration range, certain assumptions need to be made. The initial uncertainty can be calculated as $U_0 = c_0 - a_0$. As the remaining work progression reaches r_t at a time t, the current uncertainty is calculated as $U = r_t U_0$ based on the linear reduction assumption mentioned above. Then, incorporating fixed shape parameter assumption and equation (12), we can formulate the duration bound as follow:.

$$\begin{aligned} a_t &= b_t + \frac{1-\alpha}{4} r_t U_0, \\ c_t &= b_t + \frac{5-\alpha}{4} r_t U_0, \end{aligned} \tag{17}$$

where a_t and c_t are the optimistic and pessimistic duration estimation correspondingly.

The process mentioned above is shown in Figure 5.

5. System Implementation and Validation

This section presents a novel data-driven project schedule management system that combines data analysis algorithms with the conventional project management theory for unleashing the potential of computer-assisted assessments of overdue risks and criticalities. This section will introduce the overall architecture and core modules of our system. Besides, some social network analysis- (SNA-) based network indices are introduced to measure the effectiveness of our system usage.

5.1. Overall Architecture. The implementation of our BIM-based project progress management system aimed to gather progress data, evaluate current progress, and render critical tasks in a visual format. This system is a composition of the following five modules: "WBS creator," "scheduler modeler," "progress collector," "risk analyzer," and "criticality visualizer." A depiction of this acclaimed architecture is shown in Figure 6.

(i) The "WBS creator" is designed to establish a standardized WBS and automatically assign codes to tasks based on their work content. These codes are then used to generate connections between tasks and their corresponding BIM components.

(ii) The "schedule modeler" generates or modifies a schedule by defining the planned start or end of a task and its dependencies. This is accomplished using graphical programming.

(iii) The "progress collector" is responsible for collecting progress reports from the manager and updating related progress parameters automatically to provide a corrected distribution for the analyzer.

(iv) The "risk analyser" conducts reliability analysis to determine the most probable failure condition upon which the criticality visualization is based.

(v) The "criticality visualizer" visualizes the criticality of each task based on its free float. Critical tasks are highlighted for easy identification.

The suite of modules aligns with the four stages of the PDCA theory. During the "plan" phase, BIM data are collected to calculate the workload and duration of each task. Throughout the "do" stage, real-time progress information is gathered to update the distribution of each task's duration. In the "check" phase, burn-out and reliability analyses are executed to compare the deviation of actual progress and assess the risk of overdue tasks. Finally, the "action" stage applies risk visualization to spotlight the most critical tasks in the project and encourage managers to take necessary actions.

5.2. Data Collection. The following are the four types of inputs required in our system:

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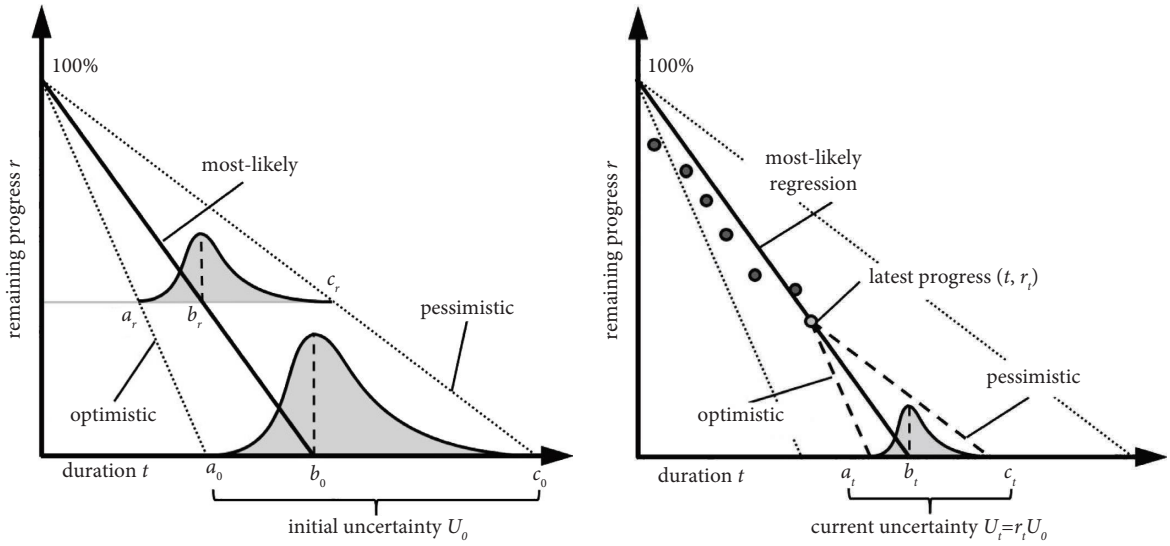


Figure 5: Illustration of approximated parameter estimation. (a) Uncertainty propagation. (b) Distribution estimation.

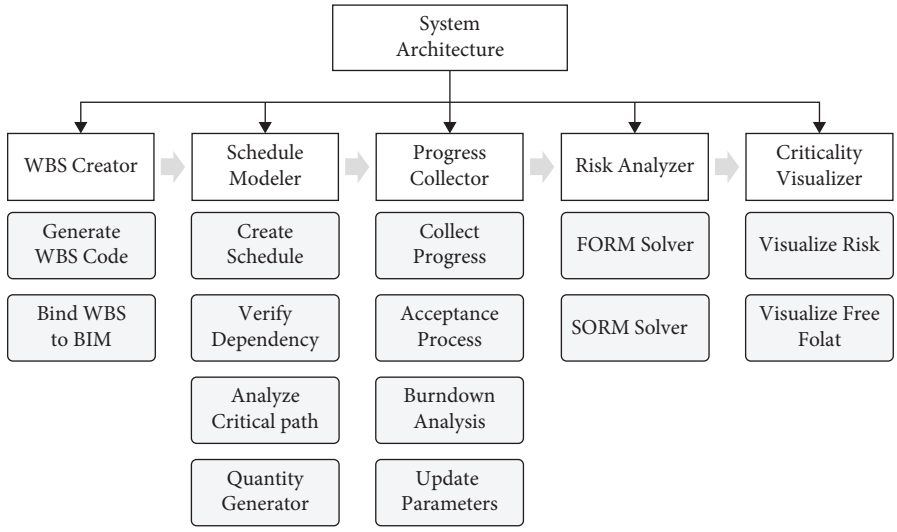


Figure 6: System architecture.

(i) Work breakdown structure (WBS), which identifies the hierarchical affiliation of each task. WBS determines how the workload is divided and how the total project progress is determined

(ii) Construction schedule network, which identifies the dependencies between each task. These dependencies are used in the critical path method to calculate the float time of each task

(iii) BIM components, including their types, materials, process types, geometric parameters, and task affiliations. The BIM data is essential for estimating workload because of the given time-consuming nature of manual methods. It lays a foundation for automatic quantity take-off

(iv) Task duration quotas for different types of labor work, for example, earth excavation, reinforcement binding, and concrete pouring. Quotas convert the workload of a task to its duration and form the foundation for automatic scheduling.

The proposed framework utilizes a MySQL Server to store plan and actual progress data from both the design and construction phases of the project. In addition, data exchange between our system and the BIM Cloud Platform happens via Web Service API, which enables BIM data and additive coloring of components. The structure of our database and the data exchange is shown in Figure 7.

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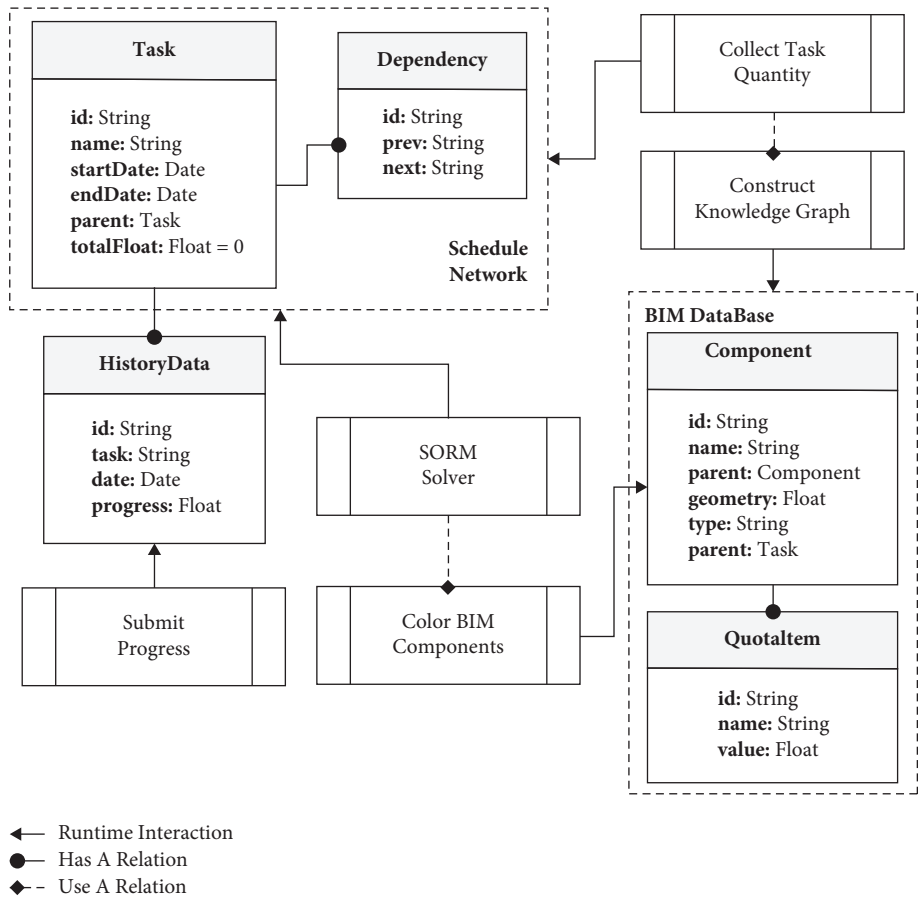


Figure 7: Structure of the database and data exchange in our system.

5.3. Integrated Functionalities

5.3.1. Integration of BIM Quantities and WBS. In order to assist project managers with preliminary estimations, quantities can be utilized as a reliable reference. Typically, experienced project managers examine and measure detailed drawings in order to approximate task durations. However, this process can be tedious and time-consuming, particularly when adjustments are made to the initial design, requiring a recalculation of the related quantities. By establishing an automated system for estimating task durations, efficiency can be improved, thus saving project managers' valuable time and effort.

Our system design employs the entity-relationship (E-R) representation to effectively model duration estimations. The components of the model are linked by a specialized relationship, which stores the corresponding quantity values. This is achieved through the use of predefined templates, which enable the instantiation or modification of parts and their associated parameters. As changes are made to a particular template, quantities and bill items are automatically recomputed, and connections are regenerated. These connections are then efficiently indexed by the search engine, which effectively sums up the quantities according to their respective working terms, as shown in Table 2.

Following the computation of quantities, task durations are determined by applying the appropriate working rate quota, which may be derived from national standards or expert experiences. These computed durations serve as reference values in the scheduling process. In addition, any design changes are identified as modification events by the search engine, which initiates updates to all indexed documents involved in the process. This mechanism ensures that the latest quantities and duration reference values are always utilized. The overall process is shown in Figure 8.

5.3.2. Progress Data Collection and Distribution Updating. Throughout the project execution, it is necessary to document all progress updates for several reasons. First, historical records serve as the foundation for addressing future progress disputes that may arise. Second, timely identification of progress deviations from historical data enables the root cause to be identified which are then resolved promptly. Third, historical records provide a reliable basis for the creation of a progress quota database, which can guide the execution of future construction projects. By meticulously documenting progress updates, construction teams can enhance project delivery and management quality.

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Table 2: Required quantity for each object and work.

Category	Object	Components	Work type	Quantity type
Earth work	Earth block	—	Site leveling Earth excavation	Area Volume
Reinforced concrete	Foundation, beam, wall, column, and slab	Rebar Formwork Concrete	Steel bar binding Formwork construction Concrete placement	Length Area Volume

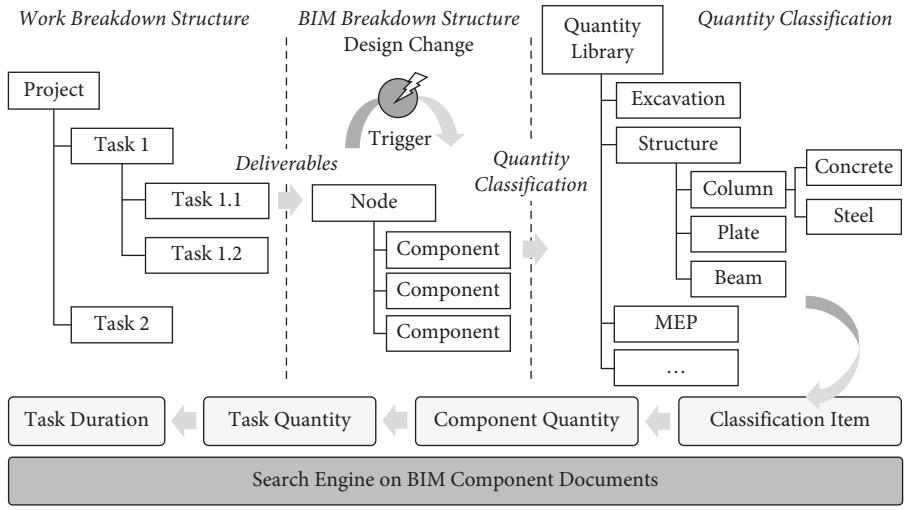


Figure 8: Illustration of integration of BIM quantities and WBS.

In our design, the progress report includes the following three key elements: "work scope," "work progress," and "acceptance certificate of progress." "Work scope" refers to the scope of the reported workload, which corresponds to a task in the WBS. By specifying the work scope, progress updates could be recursively propagated to their parent node, leading to automatic progress updates in total progress. The "work progress" refers to the complete degree of the workload. Each submission of the workload will generate an acceptance process, which will be submitted to the supervisor for acceptance. Only the reported and accepted workload can be recorded. This process reduces the nonstandard operation of the construction company to a certain extent and increases the management's ability to control the project's progress.

Once progress updates are reported, our system executes task duration estimations to update its parameters automatically. The estimator eliminates the need for manual work and offers our system the ability to handle dynamic assessments easily. The updated distributions are subsequently used in the dynamic analysis mentioned in Section 3, resulting in an accurate evaluation of the project's progress. Figure 9 shows the process of how progress updates and task duration estimations are integrated seamlessly into our project management system.

5.3.3. Implementation of Risk Visualization. Our risk analysis method relies on the estimated distributions outlined earlier in this section. Once the analysis is carried out, the distribution parameters of each task are retrieved from the database and transformed into normal space. The first-order reliability method (FORM) is employed to identify the most probable point (MPP). Then, the numerical result is derived from the SORM-FOE [24]. We then compare the failure probability given by SORM with a preset threshold, such as 5%. By doing so, we can accurately quantify the risk of a project delay. If the failure probability is higher than the threshold, it indicates that the risk is out of control, and immediate action should be taken to mitigate the risk. Conversely, if the risk is within acceptable limits, no action is required.

Once the MPP is identified, it serves as a representative case to assist the project managers in controlling the project. First, MDA is applied to MPP to calculate the free float of each task. Based on the free float, the criticality of each task can be categorized into different levels. For instance, tasks with a free float of less than 3 days are considered critical, those with a free float of less than 7 days are deemed important, and others are categorized as normal tasks. These levels can be adjusted to suit the preferences of the project manager if is necessary. Once the criticality level of each task has been established, the BIM components associated with these tasks are color-coded by the system to reflect their level of importance. This process is shown in Figure 10.

5.4. System Measurements. To assess the effectiveness of our system, we employed an egocentric network analysis, a social network analysis (SNA) technique, which enables us to

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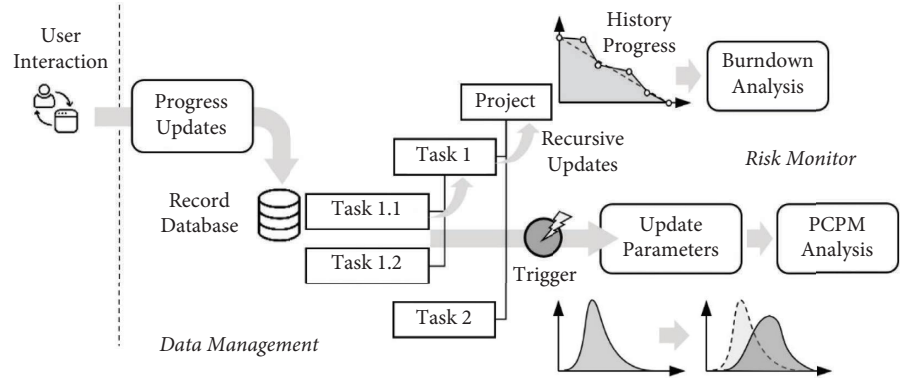


Figure 9: Illustration of progress data collection and distribution updates.

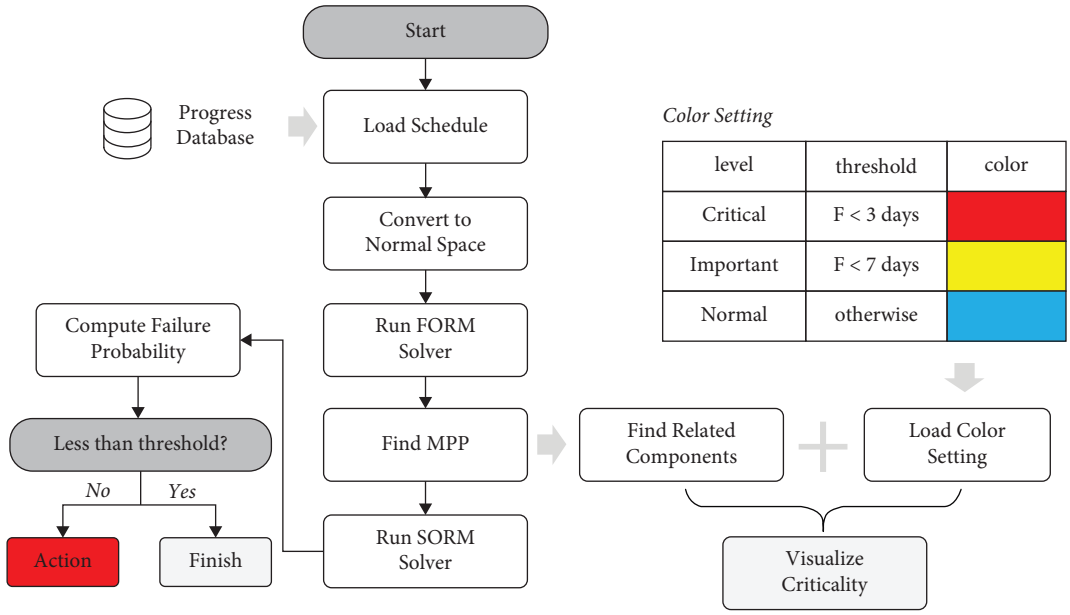


Figure 10: Illustration of progress risk visualization.

scrutinize an individual's actions and assess how their social connections impact each other [32]. We leveraged this technique to model the management team using social network models, establish changes in information exchange efficiency between different projects before and after adopting our system, and employ social network analysis methods to gauge the system's actual effectiveness.

Assuming that a project team can be conceptualized as a social network, where N represents all the members within the network and E denotes their interactions. Quantitative comparisons are possible through a set of network indicators as shown in Table 3. Employing such indicators would facilitate the assessment of the project team's collaboration status.

The node-level indicators that describe the information communication among project members are as follows: the degree centrality that represents the average communication time among members and the average distance indicating the average time of information transmission to members. In addition, network-level indicators provide an overview of the overall information communication of the project, that is, the network density indicates the degree of full communication between members, average degree indicates the average communication time of members, network diameter represents the longest information propagation path in the project, and the effective size indicates the degree of information exchange among members, excluding ego.

6. Case Study

The purpose of this section is to verify the feasibility and effectiveness of the proposed method by applying it to evaluate the progress of a steel structure medical laboratory in China. The project's duration is expected to be two years, spanning from January 2022 to December 2023. As an example, the progress of structural engineering is chosen to validate the accuracy and efficiency of our method.

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Table 3: Selected network indices for evaluation.

Level	Metrics	Formulation	Definition
Node-level	degree centrality	$Cd(N_i) = \sum_{a_{ij}} (i \neq j)$	The number directly linked to the nodes
	Average distance	$\overline{D}(N_i) = \max_{P \in \mathbb{P}(N_i)} \|P\|$	The average length of path starting from given node
Network-level	Density	$\rho(\mathbb{G}) = 2\|E\|/\|N\|(\|N\| - 1)$	The ratio between actual edges and potential edges in a network
	Average degree	$\overline{Cd}(\mathbb{G}) = \sum_{Cd}(N_i)/\|N\|$	The average of degree centrality in the network
	Diameter	$D(\mathbb{G}) = \max_{P \in \mathbb{G}} \|P\|$	The length of longest path in network
	Effective size	$\mathrm{ES}(\mathbb{G}) = n - 2t/n$	The average nonredundancy score of all the primary alters

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6.1. Case Details. Our case schedule network comprises 27 activities and 34 work dependencies, and this project comprises two main parts: the aboveground and basement structures. To facilitate the construction process effectively, the structure is divided into several working segments. The basement is segmented into four individual working sections for each floor, while the aboveground structure is broken down into three distinct working sections for each floor. Each segment is constructed floor by floor, which creates working dependencies. Figure 11 shows an illustration of the project division and schedule network.

The activity times of each working segment are shown in Table 4.

Based on the results obtained from traditional CPM calculations, the completion of the structural phase of the project is estimated to take about 165 days, and to ensure a conservative approach, an expected end time of 168 days has been established. The next subsection will evaluate the reliability of this decision.

6.2. Numerical Comparative Experiment. To obtain a reliable conclusion, each numerical method applied in our comparison was subjected to 20 simulations to calculate the mean estimation $\widehat{p}_f$ and its average running time $\overline{t}$. In addition, the stability of the numerical results was compared by computing the variation coefficients $c_v = \sigma_{p_f}/\widehat{p}_f$. The experiments were carried out on a PC equipped with an 8-core Intel Core i7-0700K 3.6 GHz CPU, and the programs were developed in Python 3.7.3 and NumPy 1.20.1. Further details on each method's settings are shownin Table 5.

The findings in the computation results, as shown in Table 6, indicate that the iterative search methods, such as LUBE, FCPM, FORM, and SORM, generally outperform the sampling-based methods of DMCS and subset simulation in terms of computational speed. This outcome may be attributed to the iterative search's singular evaluation of the schedule per iteration and conclusion within a few iterations, while the sampling-based method requires tens of thousands of schedule evaluations. Therefore, it can be concluded that the iterative search methods are a more efficient approach to scheduling problems from a computational perspective.

As for the accuracy of various methods for solving the schedule failure problem, the simulation results reveals that, while SORM is a second-order approximation to the schedule failure problem, it demonstrates accuracy that is almost equivalent to Monte Carlo simulation. Conversely, FORM has limitations as it fails to produce satisfactory results when the failure domain is highly nonlinear. Utilizing only a first-order expansion leads to an inadequate approximation of the failure domain. Consequently, the second-order reliability method that integrates a softmax function approximation could deliver a dependable solution in a relatively shorter period. These findings demonstrate that the second-order reliability method could be a promising approach to solving schedule failure problems, particularly when time constraints are an issue.

By drawing and examining the relationship between the completion time limit and failure probability, it can be observed from Figure 12 that the curve provided by SORM is quite similar to the ones presented by DMCS and subset simulation techniques. However, there is a noticeable systematic error in the graphs generated by LUBE and FCPM. It appears that LUBE concentrates specifically on the most critical path, potentially leading to a lower failure probability as other paths could also contribute to the failure, and it is also important to note that fuzzy numbers and operations may not fully capture the complexity of probabilistic modeling, resulting in some deviation.

An interesting observation is that the result obtained by the FORM is found to be similar to the one obtained by LUBE, indicating a potential similarity between their mechanisms. A possible explanation for this observation could be the similarity in the approach they take. The LUBE algorithm evaluates only the most significant path, which means that the total duration could be expressed as the sum of all the task durations on that path. In the standard normal space, the failure boundary of LUBE takes the form of a plane, much like that of FORM. However, LUBE overlooks some noncritical activities, thus resulting in small differences between the two methods.

In addition, we found that SORM consistently provides lower estimates than DMCS, which is not entirely unexpected. This is due to the fact that the softmax-based modified Dijkstra algorithm gives a duration which is less than the actual duration resulting in an underestimation of the failure probability by SORM. Nonetheless, SORM evaluates the entire network rather than just a single path, leading to a smaller deviation in its approximation when compared to DMCS. Therefore, in most cases, SORM presents an ideal balance between computational efficiency and acceptable accuracy for the schedule failure problem. However, it is advisable to exercise caution when assessing an event with a rare probability.

6.3. System Usage Efficiency Verification. To assess the practical application of the system, we administered a questionnaire to gather information on any shifts in the workflow before and after the system was applied. We then quantified the system's impact using the network indicators specified in Section 5.4. The findings are shown in Table 7. The "ego" in the ego-network below represents the project manager and is denoted by the blue node. The representatives of each specialization (structure, MEP, and curtain wall) are colored in orange, while the labor team leaders are depicted by the grey nodes. The edges connecting the nodes represent the interactions between two individuals, with the number assigned reflecting the average communication time.

At the ego-level, it can be inferred that the project manager spends less time on communication with the adoption of the proposed system since the degree of centrality indicates the average communication time for him. Moreover, the average distance from ego illustrates the time it takes to collect information from onsite work teams. By

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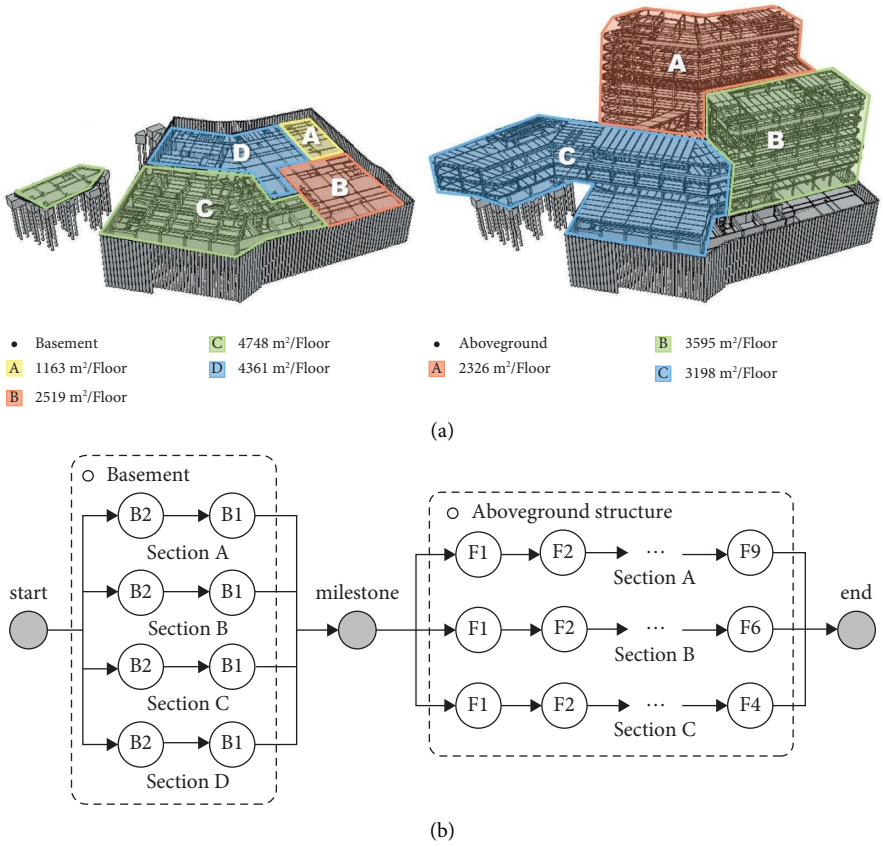


Figure 11: A schedule network of a real project. (a) Illustration of working segments: each floor is divided into several working segments and each segment is constructed floor by floor. (b) Illustration of a project schedule: there are three special nodes which are a starting point, an ending point, and a midway milestone. Their durations are assigned with zero-day.

Table 4: Statistical parameters of each work area.

	Segment	Average building area (m^2/floor)	Working days (day/floor)			#Floors
			Optimistic	Most likely	Pessimistic	
Basement	A	1163	15	17	23	2
	B	2519	17	19	25	2
	C	4748	23	24	27	2
	D	4361	22	24	26	2
Aboveground	A	2326	11	13	14	9
	B	3585	13	15	17	6
	C	3198	12	13	15	4

Table 5: Setting of each algorithm.

Methods	Setting
Direct MCS	Samples: 50,000
Subset simulation	Batch size: 5000; selection ratio: 0.1
FORM/SORM	Finite difference step: 0.01

applying our system, the information passing time has significantly decreased, resulting in a remarkable improvement in the overall efficiency of the decision-making process for project managers.

At the project level, higher network density and effective size imply a greater number of information exchanges amongst team members, while a lower average degree indicates less time spent on communication activities. Furthermore, a smaller diameter signifies that the time taken to transmit information from one end of the network to another is reduced. Ultimately, the implementation of our system has led to an improvement in the efficiency and effectiveness of information sharing, streamline information flow, and significantly reduced communication time, resulting in a positive overall impact on project management activities. In conclusion, our social network analysis found that the adoption of our system positively affects the network

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Table 6: Schedule network failure evaluation (deadline = 168 days).

Methods	Failure probability (%)	Variation coefficient	Running time (ms)
LUBE	1.52	—	0.3 ± 0.5
Triangular fuzzy CPM	27.64	—	0.7 ± 0.5
Direct MCS ∗ (baseline)	1.83	2.46%	2914.8 ± 49.2
Subset simulation	1.74	5.63%	751.5 ± 32.7
FORM	1.57	—	0.7 ± 0.6
SORM (ours)	1.75	—	1.5 ± 0.7

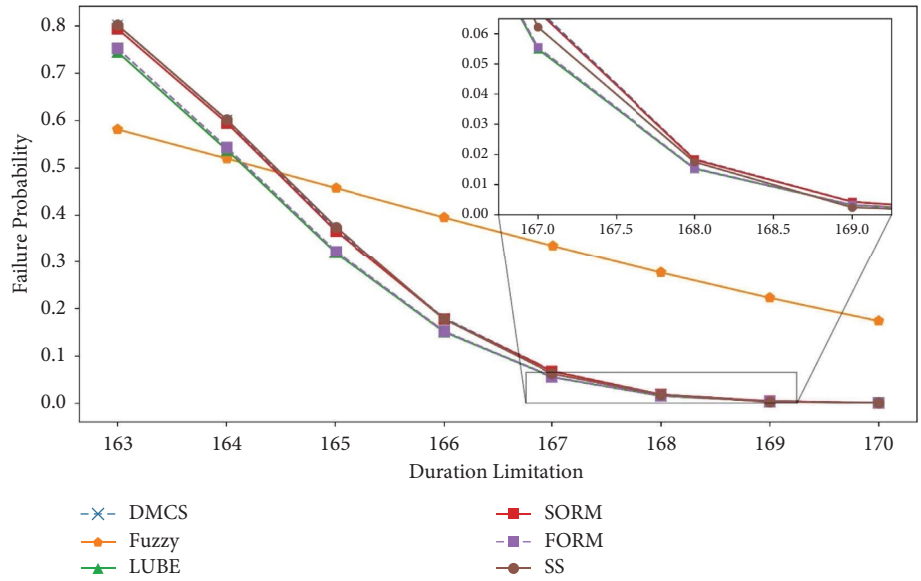
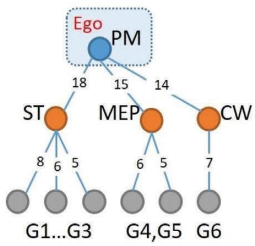
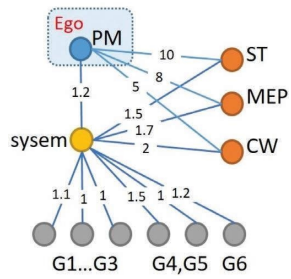


Figure 12: Failure probability curve.

Table 7: System indices in the case project.

		Without system	With system
Graphical presentation			
Ego-level	Degree centrality	15.67	6.05 (−61.4%)
	Average distance	22.5	3.82 (−83.0%)
Network-level	Density	10%	11.82% (+18.2%)
	Average degree	8.11	2.57 (−68.3%)
	Diameter	26	10 (−61.5%)
	Effective size	7.67	8.2 (+6.9%)

structure of project management activities, enabling more frequent information exchanges and leading to a reduction in time consumption, ultimately increasing the chances of project success.

7. Discussions and Conclusions

7.1. Discussions. With a novel focus on utilizing a monitor system, our approach to BIM progress management systems

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diverges from the current research by enabling the acquisition of real-time onsite progress reports, continuous parameter updates, and ongoing assessment of project progress risks. This combination grants an exceptional opportunity for real-time decision-making in project management, enabling managers to make informed choices on project requirements, budgets, timelines, and more, ensuring the successful outcome of the project in real-time.

Our case study has shown that our proposed system is highly effective and practical in both numerical computations and real-world applications. Through the case study presented above, we have demonstrated that our approach is both efficient and accurate, delivering highly reliable results. While some assumptions have been made, these were performed so judiciously, and our experimental results have confirmed that any systematic errors introduced are insignificant and can be safely disregarded for optimal accuracy. In addition, social network analysis has confirmed that our system is highly practical and useful in practical settings, enabling project managers to streamline their work and reduce the incidence of errors.

Our main contributions are as follows:

(1) We put forth an innovative implementation of the probabilistic critical path method (PCPM), by integrating the second-order reliability method (SORM) and softmax approximation, which demonstrates a higher degree of efficiency than conventional sampling-based techniques while maintaining equivalent levels of accuracy

(2) We introduce a novel method for estimating project duration distribution utilizing the beta PERT-Beta distribution model and three reasonable assumptions. The approach is effective and practical, with significant value in the field of duration estimation

(3) Our study offers a comprehensive solution that spans the entire project lifecycle, from scheduling and execution to monitoring and closure, providing a more holistic approach to project management

(4) Our research introduces a novel method that employs a search engine to automatically index Building Information Modeling (BIM) data, calculate a precise quantity list, and prepare a seamless plan, significantly fastening the planning process

However, there are still some limitations of our method which are as follows:

(1) One of the significant challenges that our analysis and visualization system faces is the initial setup process. The creation of a quota database is a complex task that requires extensive experience and attention to detail. The hierarchical structure of quota items must be carefully organized, and the quota values must be assigned with great care. In order to ensure the accuracy and reliability of the quota values, we engaged the expertise of construction management professionals, who were invited to complete a survey as part of this study. Despite the extensive work required to establish the quota database, the benefits of the system once it is set up are significant and well worth the initial investment of time and effort.

(2) Not all the work could be properly modeled in BIM. For instance, paint coating cannot be directly incorporated into the BIM model due to its complexity, which may make the model unnecessarily complicated and difficult to use. In such cases, it is essential to extract the paint coating information from the area property of the component it belongs to. This requirement underscores the importance of defining the properties of the component family effectively, so that the data it conveys covers all the relevant workloads. This situation presents considerable challenges for BIM modeling and demands advanced skills from modeling personnel to come up with accurate and comprehensive models while maintaining simplicity and ease of use.

(3) Our proposed method may not be suitable for linear projects, because linear projects are typically modeled by the linear schedule method (LSM), which is a different approach than what CPM used in our proposed method. Nevertheless, our proposed method can still be a viable option through the technique of discretization. By breaking down the linear project into smaller segments, our method can be adapted to the linear project.

(4) Our system's effectiveness is affected by the requirement of manual progress reports, which requires time and is prone to errors. However, the system's automated data analysis and visualization tools enhance project management efficiency, accuracy, and overall performance, outweighing the drawbacks of manual reporting. In summary, our system significantly reduces the workload of manual reporting by providing real-time information on project progress, risks, and opportunities while eliminating errors and enhancing the overall performance.

7.2. Conclusions. Conventional methods of managing construction project schedules have inherent limitations, largely due to their heavy reliance on human efforts. As an improvement, our study proposes a SORM-based data-driven progress evaluation algorithm, supplemented by an analysis and visualization system. Through the implementation and use of this approach, we find that it can be a powerful and straightforward tool for managing project progress. The results of our testing and analysis can be summarized as follows:

(1) This study offers a promising approach by introducing an approximation-based second-order reliability method. By utilizing the softmax function to replace the max function in the modified Dijkstra algorithm, we were able to differentiate the total duration. While this approach may carry some

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systematic errors, our numerical experiment demonstrates that these errors are typically insignificant. In comparison to Monte Carlo Simulation, our method is not only faster, but it retains a high degree of accuracy. This advantage is particularly notable when dealing with large schedule networks, where speed and accuracy are of the utmost importance.

(2) In our study, we present a novel approach for estimating duration distributions based on historical data, utilizing the beta PERT-Beta distribution as the probabilistic model. To simplify the problem, we make three reasonable assumptions, which allow us to achieve highly efficient results. Our simplified method has been proven effective and practical in real-world applications, demonstrating the significant value it brings to the field of duration estimation.

(3) We present a novel progress management system that enables the quantitative analysis of overdue completion risk of schedules. The proposed system combines BIM and reliability theory to harness the power of BIM information in providing valuable reference values for duration estimation. Moreover, BIM plays a pivotal role in visualizing the analysis results to facilitate better interpretation and comprehension of the criticality of the data. To validate the system's feasibility, we undertook a real construction project case study, which entailed a 27-activity schedule as a test case, and utilized SNA results. The findings of our study demonstrated that the system markedly expedited the reporting and monitoring process while greatly improving user comprehension of the analysis results through BIM visualization and smooth information exchange between members.

Data Availability

All the data used in our numerical experiment are included in the paper.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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